RK-A-WRS · REV 3 · 05 SEP 2026 · TERRACE RESERVOIR, GRAVITY FEED

Closed-Loop Water Recovery

Both reservoirs move to the terrace, one floor above the room. That single decision deletes the supply pump, lets every bed flood by gravity, and turns the recovery side into one trough, one pump and one rising main. It also introduces three risks that did not exist before, and they are the most important part of this document.

CIRCULATED	MAKE-UP	RECOVERY RATIO	DISCHARGE CUT
1,914 L/day	**235 L/day**	**87.7%**	**554 m^3/yr**

PUMPS IN SYSTEM	SKID COST
1 duty + 1 standby	**₹79k / ₹153k**

---

01 · THE TERRACE DECISION

What moving the tanks upstairs actually buys

Rev 2 had a floor-standing 500 L reservoir, a supply pump, a catch trough, a transfer pump, a recovery reservoir and a return pump. Putting the reservoir on the terrace collapses most of that.

ELEMENT	REV 2 — PLANT IN THE ROOM	REV 3 — RESERVOIR ON THE TERRACE
Supply pump P-01	0.55 kW, 30 L/min @ 18 m, runs every flood	DELETED Gravity does it. 2.66 m of head at the top tier against 0.33 m of loss
Recovery reservoir TK-03	300 L at floor level, plus a return pump	DELETED The gate is now decided in the trough, before anything is pumped
Return pump P-03	0.25 kW	DELETED

Transfer pump P-02	0.37 kW, 25 L/min @ 12 m	Uprated to **0.75 kW, 25 L/min @ 20 m** — it now lifts to the terrace as well as pushing through the filters
Room floor area for plant	5.73 m² bay	**1.02 m² service corner** — one rack instead of three
Racks in the room	9	**11** · grow area 29.0 m² · floor multiplier **1.30×**
Fill concurrency	One tier at a time, pump-limited	Up to four tiers at once — gravity has the head to spare. Room cycle 88 min → 22 min
Carrying fertiliser salts	To the room	**Not to the roof** — dosing injects into the rising main at floor level

> **The dosing point is the detail that makes this work.** The obvious objection to a roof tank is that someone has to carry 25 kg bags of A and B salts up a ladder three times a week. They don't: the dosing skid stays at floor level and injects into the *rising* main, so every litre of recovered water is dosed on its way up and mixed by its own inflow into the tank. EC in TK-01 is the feedback signal. The only thing that ever goes up there is water.

---

02 · GRAVITY FEED

Does 3.75 m of terrace actually flood a bed at 1.5 m?

Comfortably, and with enough spare head to flood all four tiers at once. The arithmetic matters because it is what deletes the supply pump.

```
terrace slab +3750 mm above room floor
tank on a 500 mm plinth → outlet at +4250 mm, worst case is a nearly-empty tank
top tier fill nozzle +1590 mm
available static head = 4250 - 1590 = 2.66 m

losses at the design fill rate of 7.2 L/min per tier (Hazen-Williams, C = 150):
 DN32 down-main, 14 m 0.028 m
 DN25 riser, 1.5 m 0.012 m
 DN20 tier drop, 1.0 m 0.023 m
 DN20 fill solenoid, Kv 4 0.119 m ← the dominant loss
 fittings 0.150 m
 total 0.33 m → margin 2.33 m · passes by a factor of eight
```

```
with all four tiers of a rack filling together, 28.8 L/min in the down-main:
 total 0.67 m → margin 1.99 m · still passes

head runs out at roughly 15 L/min per tier, twice the design rate
```

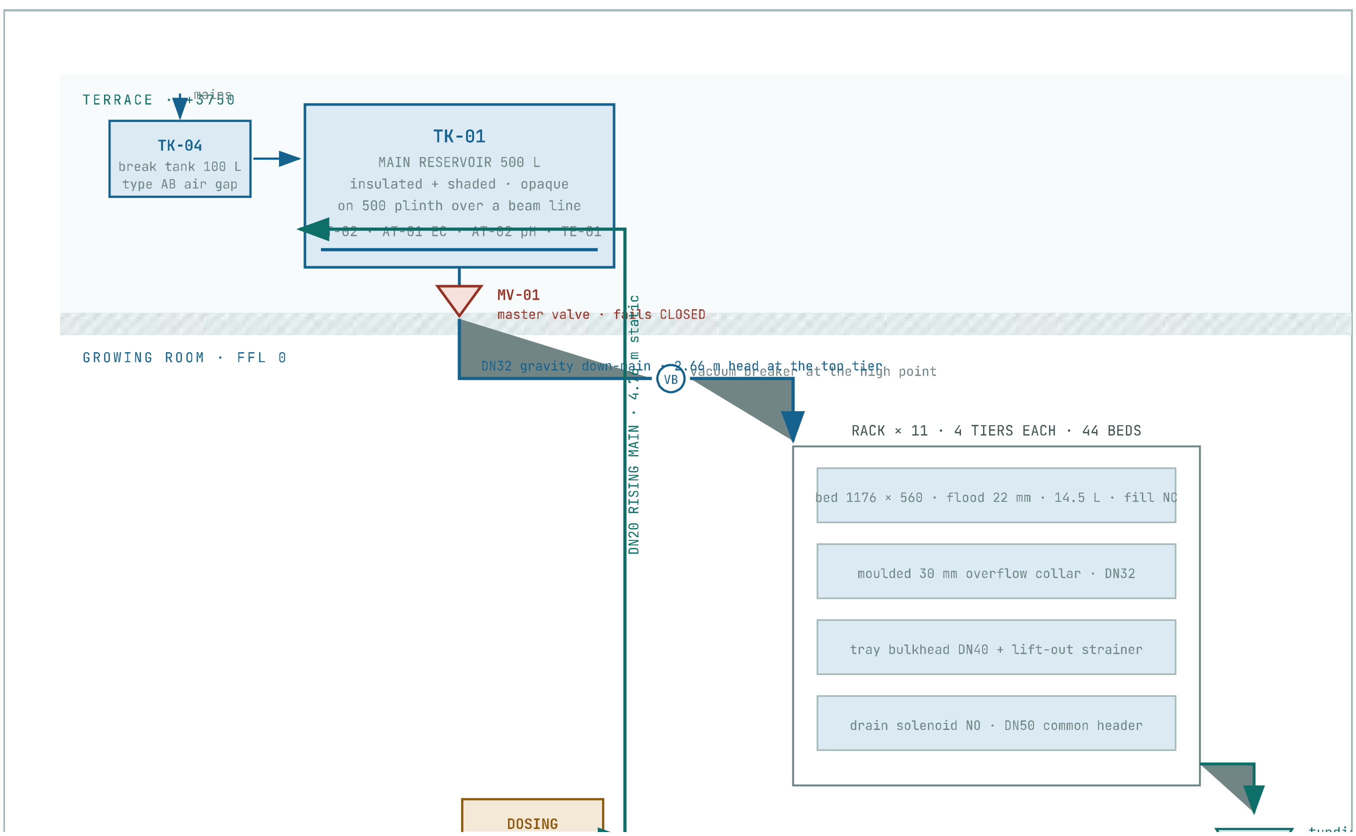
Two consequences follow. First, there is no supply pump anywhere in the system — the only pump is the one lifting recovered water back up. Second, the sequential-filling rule that existed to protect a 7.2 L/min pump is no longer needed on the supply side. **A rack can flood all four tiers at once**, which takes the whole-room fill window for eleven racks from 88 minutes to about 22.

> **Sequential *draining* is still mandatory.** Nothing about the terrace changes the DN50 drain stack or the size of the catch trough. Draining stays one tier at a time, and — new in this revision — one *rack* per recovery batch, because the gate is decided on a settled batch and the trough holds 79 L against a 46 L rack drain.

---

03 · PROCESS

The loop, end to end



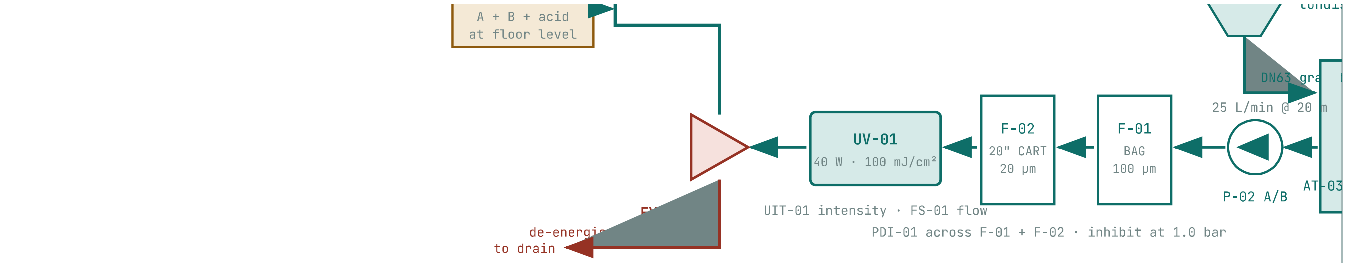


FIG 1 — REV 3 PROCESS FLOW. BLUE IS GRAVITY, TEAL IS PUMPED RECOVERY, AMBER IS DOSING, RED IS THE WASTE PATH AND THE FAIL-SAFE POSITIONS. ONE PUMP, ONE RISING MAIN, ONE CROSSING OF THE CLEAN/UNVERIFIED BOUNDARY AT FV-01.

04 · NEW RISKS

Three things that did not exist when the tanks were indoors

This is the part of the terrace decision that has to be engineered rather than assumed. All three are manageable; none is optional.

Risk 1 — solution temperature

A tank on a terrace is a solar collector. In Coimbatore, an unshaded HDPE tank routinely reaches 40 °C by mid-afternoon in April and May. Nutrient solution above 26 °C loses dissolved oxygen and becomes hospitable to *Pythium* — it is gate G3 in the reuse logic, and here it would be failing on the *supply* side, where there is no gate at all.

dissolved O_2 at 20 °C **9.1 mg/L**
dissolved O_2 at 30 °C **7.6 mg/L** — a 16% loss before any root has used a molecule
dissolved O_2 at 40 °C **6.4 mg/L** — **30% loss, and Pythium zoospore activity peaks in this band**

Minimum — Ooty

Opaque tank under a shade structure, 25 mm PUF wrap with white UPVC cladding, and fill/draw scheduled so the tank turns over overnight. Ooty's ambient does the rest. **Sufficient for the first six months.**

Required — Coimbatore

Everything above, plus temperature logged continuously with an alarm at 26 °C, and budget for a ½ TR inline water chiller. Assume you will need it; be pleased if the data says otherwise.

Risk 2 — a 500 L tank with a gravity path into the room

A burst hose or a failed union downstairs, with the tank full, empties 500 litres onto the growing room floor. Nothing in the Rev 2 architecture could do that, because the reservoir sat on the same floor as the leak.

- **MV-01, motorised master valve at the tank outlet, spring-return closed.** Energised only during a flood window. Everything downstream of it is dry between floods.
- **Vacuum breaker at the down-main high point**, so a downstream failure cannot siphon the tank.
- **Room flood sensor de-energises MV-01 directly**, hard-wired, not through the bus.
- **Flow totaliser on the down-main.** A flood window that delivers more than 1.3 × the expected volume closes MV-01 and alarms — this catches a slow leak that a flood sensor at one floor point would miss.

Risk 3 — terrace slab loading

```
TK-01 filled 500 L + tank 25 kg + plinth ≈ 560 kg
TK-04 filled 100 L + tank ≈ 110 kg
on a 1.0 m² footprint = 5.5 kN/m² — an accessible terrace is typically designed for 1.5–2.0

spread over a 2.0 × 1.5 m platform bearing on beam lines:
 670 kg / 3.0 m² = 2.2 kN/m² — at the limit, and only acceptable over beams, never mid-span
```

> **This needs a structural engineer's sign-off before the tank is ordered, not after.** The load is concentrated, permanent and sits on a slab that was probably designed for foot traffic. The platform must bear on the beam or column lines, and the check has to be done against the building's actual drawings. It is the one item in this document that cannot be resolved by choosing better equipment.

05 · WATER BALANCE

Eleven racks, 44 beds, three cycles a day

STREAM	L / CYCLE	L / DAY	M³ / YR	BASIS
Gross flood delivered	638	1,914	632	44 beds × 14.5 L (1176 × 560 × 22 mm)
Evapotranspiration	14.5	43.5	14.4	1.5 $L \cdot m^{-2} \cdot d^{-1}$ at DLI 10.4 × 29.0 m^2 canopy
Water leaving with harvest	6.3	19.0	6.3	4.8 kg fresh per bed, 90% water, 10-day cycle
Substrate charge on re-sowing	8.8	26.4	8.7	6 L per bed per 10-day cycle
Returned to trough	608	1,825	602	gross − consumption
Blowdown for salt control	48.7	146	48.2	8% of returned volume, or as Na^+ demands
Make-up required	78.3	235	77.6	**consumption 88.9 + blowdown 146**

```
recovery ratio = 1 - 235 / 1,914 = 87.7%
annual discharge, closed loop 48.2 m³
annual discharge, drain-to-waste 632 m³
reduction = 554 m³/yr · 92%
```

One rack per recovery batch. The trough holds 79 L of working volume against a full rack drain of 46 L, and the reuse gate is decided on a settled, mixed batch rather than a moving stream. That sets the cadence: drain one rack (2.7 min), settle (30 s), test (3 min), transfer (2 min) — about 8 minutes per rack, 88 minutes for eleven, three times a day. It is the drain side, not the fill side, that now sets the room's clock.

06 · EQUIPMENT

What is left, after the terrace deleted three items

TAG	ITEM	LOCATION	DUTY / SPECIFICATION	WHY
TK-	Main nutrient	Terrace	500 L opaque HDPE, 25 mm	Opaque against algae,

01	reservoir		PUF + white cladding, shaded, on a 2.0 × 1.5 m spread platform	insulated against the sun, spread against the slab
TK-04	Make-up break tank	Terrace	100 L, type AB air gap, float valve, 5 µm + carbon pre-filter	Backflow prevention belongs at the highest point in the system
MV-01	Master supply valve	Terrace, tank outlet	DN32 motorised, **spring-return closed**	The only thing standing between 500 L and the room
VB-01	Vacuum breaker	Down-main high point	DN32 anti-siphon	A downstream failure must not siphon the tank
TK-02	Catch trough	Room, NE service corner	1400 × 650 × 200 mm fabricated HDPE, inlet at 87 mm, 79 L working, bunded lip	Shallow because tier 1 discharges at 170 mm; a standard drum cannot be gravity-fed
S-01	Basket strainer	In TK-02	200 µm SS316, lift-out	Root hairs and coir fines, before they reach an impeller
P-02 A/B	Transfer / lift pump, duty + standby	Room, service corner	**0.75 kW, 25 L/min @ 20 m**, self-priming, EPDM seals, IP55, auto-changeover	One pump now does filtration, disinfection and the 4.76 m lift. Standby because there is no second path
F-01	Bag filter	Skid frame over TK-02	100 µm, size-1 bag, 3 spares	Bulk solids; cheaper per gram of dirt than a cartridge
F-02	Cartridge filter	Skid frame	20-inch wound PP, 20 µm, 5 spares	20-inch, not 10-inch — a single 10" element is rated 11–15 L/min and would throttle P-02
UV-01	UV-C disinfection	Skid frame	40 W amalgam, 316L, quartz sleeve, intensity sensor · 1.5 m^3/h at 100 mJ/cm^2	100 mJ/cm^2 is the horticultural design dose for oomycete and bacterial control
FV-01	Diverter valve	Skid frame, before the rising main	DN25 motorised 3-way, **spring-return to waste**	Unverified water never enters the rising main without power and a

				passed gate
DOS-01	Dosing skid	Room, floor level	A + B + acid peristaltic, injecting into the rising main	Nobody carries fertiliser salts to a roof

```
P-02 duty check
static lift, trough water 87 mm → tank inlet 4850 mm 4.76 m
DN20 rising main, 14 m at 25 L/min 1.13 m
F-01 + F-02 clean 3.70 m
UV-01 2.80 m
fittings and the diverter 0.60 m
clean duty 12.4 m · at the 1.0 bar filter-change limit 18.9 m
specify 25 L/min at 20 m — 0.75 kW, not the 0.37 kW of Rev 2
```

07 · INSTRUMENTS AND THE GATE

The reuse decision moves upstream

In Rev 2 the water was filtered, disinfected and stored, and only then tested. In Rev 3 it is tested first, in the trough, and only water that passes is ever pumped. That saves a tank, a pump and the energy of lifting water you are about to throw away.

GATE	CONDITION	THRESHOLD	MEASURED AT	ON FAILURE
G1	Electrical conductivity	≤ 1.5 × setpoint — ≤ 1.80 mS/cm at a 1.20 setpoint	AT-03, in the trough	Trough dump valve to waste
G2	pH	4.5 – 7.5	AT-04, in the trough	Dump to waste
G3	Temperature	≤ 26 °C	TE-02, in the trough	Hold 60 min, re-test, then dump
G4	Filter differential pressure	< 1.0 bar across F-01 + F-02	PDI-01	Inhibit P-02, alarm, change elements
G5	Disinfection proven	UIT-01 ≥ 70% *and* lamp on *and* FS-01 made	UV-01	FV-01 to waste mid-transfer
G6	Leak detectors clear	No leak puck active this cycle	LD-01…11	Dump to waste; isolate that rack

G7	Operator contamination hold	Not set	Controller	Dump to waste until cleared by a named operator
G8	TK-01 has room	LT-02 < 90%	Terrace	Hold in trough until level falls

Sequence for one recovery batch

01 **One rack drains.** Four tiers in sequence into the trough — 46 L. LSH-04 confirms each header is empty before the next tier.

02 **Settle 30 s.** Solids to the strainer basket, air out of the sample.

03 **Test 3 min.** AT-03, AT-04 and TE-02 read as 60-second rolling means on the mixed batch, not spot values on a moving stream.

04 **Evaluate G1, G2, G3, G6, G7, G8.** Fail on any → the trough dump valve opens to waste and the event is logged with the failing gate named. Nothing is pumped.

05 **UV pre-strike.** On a pass, UV-01 energises and holds 60 s of amalgam warm-up before the pump starts.

06 **Transfer.** P-02 runs; FS-01 must make within 10 s. G4 and G5 are watched continuously — either failing throws FV-01 to waste mid-transfer.

07 **Dose on the way up.** DOS-01 injects A, B and acid into the rising main, proportional to the trough EC deficit against setpoint.

08 **Stop.** P-02 stops on LS-01 low or a 4-minute run limit — the run limit is what catches a stuck float.

09 **Blowdown.** Daily at 02:00, the first batch is dumped regardless of gate result. This is what stops sodium accumulating; it is not an error condition.

> **The supply side has no gate.** Water in TK-01 has already passed once, but it sits on a roof between uses, warming and concentrating by evaporation. TK-01 therefore carries its own EC, pH and temperature block, read before every flood window, with MV-01 inhibited if temperature

> exceeds 28 °C. That is a supply-side interlock, not a reuse gate, and it is the direct consequence of putting the tank in the sun.

---

08 · FAIL-SAFES

De-energised states

DEVICE	DE-ENERGISED	REASON
MV-01 terrace master valve	CLOSED	500 L above the room has exactly one thing holding it back, and it must not be electricity
Rack fill solenoids	CLOSED	No bed fills unattended
Rack drain solenoids	OPEN	Beds self-drain; no standing water on a dead rack
FV-01 diverter	TO WASTE	Unverified water never enters the rising main
Trough dump valve	CLOSED	A power cut must not silently discard a good batch
Make-up float valve	MECHANICAL	Float, not solenoid — TK-04 keeps working through a power cut
P-02, UV-01, DOS-01	OFF	—

Every vessel keeps a gravity overflow one nominal size larger than its largest inlet, terminating with a visible air gap: TK-01 and TK-04 overflow to the terrace rainwater outlet, TK-02 to the room's waste. An overflow discharging into a closed pipe is not an overflow — it is a second way to pressurise the tank.

> **Recirculation still spreads root disease, and the terrace does not change that.** A shared loop connects all 44 beds hydraulically; *Pythium* and *Fusarium* zoospores travel in solution and one infected bed can inoculate the room in two cycles. UV-C at ≥ 100 mJ/cm^2, 20 µm pre-filtration, and the G7 operator hold that drops the room to drain-to-waste in one button press are the three controls, and none is optional at production scale. **UV-C also degrades Fe-EDTA** — specify Fe-DTPA or Fe-EDDHA and verify iron by lab test monthly, or you will diagnose a pH problem that is actually a chelate problem, two weeks late.

09 · COST

Two build levels

ITEM	QTY	RATE ₹	R1 ₹	R2 ADD ₹
TK-01 500 L opaque HDPE + PUF insulation + cladding	1	14,200	14,200	—
Terrace platform, GI, spread over beam lines	1	11,500	11,500	—
[illegible]	1	[illegible]	[illegible]	